



আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong

Department of Computer Science and Engineering
Faculty of Science and Engineering

Project Proposal
Group E

FOOD DELIVERY TRACKING NETWORK

Simulation using OMNeT++ with INET Framework

Computer Networks Lab
Course Code: CSE-3634

Submitted to

--	--

Program	B.Sc. in Computer Science and Engineering
Semester	7th Semester
Section	7BF
Date of Submission	May 23, 2026

1. Title

Food Delivery Tracking Network

Simulation using OMNeT++ with INET Framework

2. Introduction

Modern food delivery systems such as Pathao Food, Shohoz Food, and Foodpanda rely heavily on real-time network communication between multiple entities — customers, central servers, restaurants, delivery riders, and GPS/tracking systems. This project aims to simulate the entire network communication lifecycle of a food delivery application using OMNeT++, a discrete event simulation framework.

The simulation models how order request packets travel from a customer node to a central server, get forwarded to the appropriate restaurant, and how a rider's location updates are transmitted in real time. The goal is to visualize and measure key networking behaviors such as packet delay, throughput, packet loss, queue congestion, and routing efficiency in a structured client-server topology.

3. Motivation

The motivation behind this project stems from the rapid digitalization of the food delivery industry in Bangladesh and globally. As these platforms scale to handle thousands of simultaneous orders, the underlying network infrastructure must handle significant communication load reliably and efficiently.

This project is important, relevant, and useful for the following reasons:

- Food delivery apps are a real-world, relatable domain that clearly demonstrates networking concepts such as client-server architecture, packet routing, and congestion management.
- OMNeT++ allows accurate discrete-event simulation of network behaviors that are difficult to observe in live production environments.
- Understanding network bottlenecks in delivery systems can improve application design and infrastructure planning.
- The project provides hands-on experience with INET Framework, NED language, C++ message handling, and performance metric collection — all critical skills in computer networking.

- Bangladesh's growing gig economy and food-tech sector make this simulation directly applicable to local industry scenarios.

4. Background

OMNeT++ (Objective Modular Network Testbed in C++) is an open-source, component-based, modular, and extensible simulation environment. It is widely used in academia and industry for modeling communication networks, multiprocessors, and distributed systems. OMNeT++ uses the NED (Network Description) language to define network topologies and C++ for implementing module behaviors.

The INET Framework, built on top of OMNeT++, provides standard protocol implementations including TCP/IP, UDP, wireless mobility models, and queue management modules that are directly applicable to this project.

A food delivery system's network architecture is based on the client-server model, where:

- Customer nodes act as clients generating order requests.
- A Central Server acts as the primary controller, receiving, processing, and routing orders.
- Restaurant nodes receive preparation requests and assign delivery agents.
- Rider nodes communicate location updates periodically via GPS/Tracking messages.
- A Database module stores order state and history.

Key networking concepts relevant to this project include packet transmission, routing protocols, delay analysis, queue management, congestion control, and wireless communication simulation for rider mobility.

System Architecture

Node	Role	Communication Type
Customer	Places food orders	TCP/UDP to Central Server
Central Server	Controls all routing and processing	TCP to Restaurant & Database
Restaurant	Receives and processes orders	TCP to Central Server & Rider

Rider	Delivers orders, sends GPS updates	UDP (wireless) to Tracking System
GPS/Tracking System	Receives real-time location updates	UDP from Rider
Database	Stores order records and status	Internal TCP to Server

Networking Concepts Used

Concept	Application in Project
Client-Server Model	Customer app communicates with Central Server
Packet Transmission	Order data encapsulated into packets
Routing	Server routes packets to Restaurant or Rider
Queue Management	Multiple orders queued at Server/Restaurant
Delay Analysis	Processing, propagation, and queuing delays measured
Wireless Communication	Rider mobility and GPS update simulation
Packet Loss Simulation	Random drop probability during high traffic
Congestion Control	Buffer overflow during rush-hour simulation

5. Potential Outcomes

Upon successful completion of this project, the following outcomes are expected:

5.1 Functional Simulation

- A fully functional OMNeT++ simulation of a food delivery network with at least 3 customer nodes, 1 central server, 2 restaurant nodes, and 2 rider nodes.
- Real-time animated packet flow visualized in the OMNeT++ GUI with color-coded nodes (Customer: Blue, Server: Red, Restaurant: Green, Rider: Yellow).
- A working NED topology file, C++ module implementations, and custom message types (OrderMessage, TrackingMessage, ConfirmationMessage).

5.2 Performance Metrics

Metric	Description	Expected Tool
Throughput	Number of orders successfully delivered per second	OMNeT++ Statistics

End-to-End Delay	Time from order placement to delivery confirmation	Scalar/Vector Recording
Packet Loss Rate	Percentage of dropped packets under high load	OMNeT++ Queue Module
Queue Length	Number of pending orders at Server/Restaurant	INET Queue
Server Response Time	Time for server to acknowledge and route an order	Event Timestamps

5.3 Scenario Analysis

- Normal operation: 3 customers, 1 server, 2 restaurants, 2 riders — baseline performance measured.
- Congestion scenario: 20 simultaneous customers generating orders to observe queue buildup and packet loss.
- Rush-hour simulation: Artificial delay injected during peak time intervals to observe throughput degradation.
- Packet loss scenario: 10% random packet drop introduced to simulate poor network conditions.
- Priority order scenario: VIP orders assigned higher priority in the server queue.

5.4 Deliverables

- Complete OMNeT++ project source code (NED, C++, .ini, .msg files).
- Performance graphs: Delay vs. Number of Orders, Throughput vs. Time, Queue Length vs. Load.
- Final project report with architecture diagrams, simulation results, and analysis.
- Viva preparation sheet with anticipated questions and answers.

6. Conclusion

This project on Food Delivery Tracking Network simulation using OMNeT++ provides a comprehensive, real-world application of computer networking principles. By modeling the complete communication flow of a food delivery platform — from customer order placement through server processing to rider GPS tracking and delivery confirmation — the project demonstrates core concepts including client-server architecture, packet routing, queue management, delay analysis, and wireless communication simulation.

The project is technically feasible, academically rigorous, and highly relevant to Bangladesh's growing digital economy. It is expected to produce meaningful performance insights, a

professional simulation environment, and a strong foundation for understanding real-world network system design and analysis.

7. References

- [1] Angelis, I., Tsipis, A., Christopoulou, E., & Oikonomou, K. (2024). TCP Cubic Implementation in the OMNeT++ INET Framework for SlOT Simulation Scenarios. In: CollaborateCom 2023. LNICST, vol. 561. Springer, Cham. https://doi.org/10.1007/978-3-031-54521-4_2
- [2] Manzoor, S. & Yin, Y. (2023). Which Simulator to Choose for Next Generation Wireless Network Simulations? NS-3 or OMNeT++. Engineering Proceedings, 46(1), 36. MDPI. <https://doi.org/10.3390/engproc2023046036>
- [3] Tokkozhina, U., Mataloto, B. M., Martins, A. L., & Ferreira, J. C. (2024). Decentralizing Online Food Delivery Services: A Blockchain and IoT Model for Smart Cities. Mobile Networks and Applications, 29, 59-69. Springer. <https://doi.org/10.1007/s11036-023-02119-5>
- [4] Moumen, I., Rafalia, N., Abouchabaka, J., & Aoufi, M. (2023). Real-time GPS Tracking System for IoT-Enabled Connected Vehicles. E3S Web of Conferences. EDP Sciences. <https://doi.org/10.1051/e3sconf/202346900021>
- [5] Veseky, V., & Zavrel, J. (2021). Quality Control Methodology for Simulation Models of Computer Network Protocols. arXiv preprint. <https://arxiv.org/abs/2109.12854>
- [6] INET Framework — Official Documentation & API Reference. OMNeT++ Community. (2024). <https://inet.omnetpp.org/docs/>
- [7] OMNeT++ Simulation Manual, Version 6.x. Opensim Ltd. (2023). <https://omnetpp.org/documentation/>